

TECHNICAL SPECIFICATION FOR G.I. WIRE AND G.I. STAY WIRE

1.0 SCOPE:

The specification covers manufacture, testing preferably at manufacturer's works before despatch, supply and delivery of G.I. Wire and G.I. Stranded Stay Wire.

2.0 STANDARDS:

- i) The G.I. wire shall comply with IS: 280 – 2006 with the latest amendments and galvanising shall comply with IS: 4826 – 1979 with the latest amendments.
- ii) The galvanised stranded stay wire shall comply with IS: 2141 – 2000 with the latest amendments and galvanising shall comply with IS: 4826 – 1979 with the latest amendments.

3.0 GENERAL REQUIREMENTS:

- 3.1 a) The requirements for chemical composition for the wires as per IS: 280 – 2006 shall conform to those given in IS: 7887 – 1992. Chemical composition as per the Clause No. 6 of IS: 7887 – 1992 is given below:

Chemical composition :

- i) The Ladle analysis of steel when analysed in accordance with relevant parts of IS: 228 or any other established instrumental/chemical method shall be as given below:

<u>Constituent</u>	<u>Percent</u>
Carbon	0.18 to 0.230
Manganese	0.30 to 0.60
Sulphur	0.050 Max.
Phosphorus	0.050 Max.

- ii) Product Analysis:

Permissible variation in case of product analysis from the limits specified under Clause No. 6.1 of IS: 7887 – 1992 shall be as follows:

<u>Constituent</u>	<u>Variation over specified maximum limits, percent, max.</u>
Carbon	0.02
Manganese	0.03
Sulphur	0.005
Phosphorus	0.005

- b) The stranded stay wires as per IS: 2141 – 2000 shall not contain sulphur and phosphorus exceeding 0.060 percent each.
- 3.2 All finished wires shall be circular in section and be well and clearly drawn to the dimension specified. The wire shall be round, free from splits, scale, surface flaws, rough jagged and imperfect edges and other harmful surface defects.
- 3.3 The wire shall be in continuous length. However, the joints, if reqd., should conform to IS- 2141-2000.
- 3.4 Galvanising shall be heavy as per IS: 4826 – 1979 with the latest amendments.
- 3.5 For the purpose of galvanising, the wire shall be coated with Zinc conforming to at least Grade Zn 98 specified in IS 209 – 1966. The Zinc coating shall be uniform, adherent, reasonably smooth and free from imperfections as per IS: 4826 – 1979 with the latest amendments.

4.0 SIZE AND DETAILS:

4.1 FOR GALVANISED MILD STEEL WIRE :

- 4.1.1 Diameter of G.I. Wires shall be as per the items of the Tender. Tolerances permitted on the diameter of wire shall be as per IS: 280 - 2006.
- 4.1.2 Tensile properties of the wire shall be as follows:
Tensile strength of galvanised M.S. Wire shall be 300 – 550 N/mm².
- 4.1.3 Type of coating of galvanise shall be heavy as per IS: 4826 – 1979 and the wires shall be galvanised after drawing & annealing.
- 4.1.4 Weight per metre length of the wire shall be stated by the Tenderer.

4.2 FOR GALVANISED STRANDED STAY WIRE :

- 4.2.1 Diameter of G.I. Stay Wires shall be as per the items of the Tender. Tolerances permitted on the diameter of wire shall be as per IS: 2141 – 2000.

4.2.2	Grade as per IS:2141-1979 <u>Strength or latest</u>	Size of the <u>Stay wire</u>	Minimum <u>Tensile</u>
	4	As per the items of the Tender	700 N / mm ²

- 4.2.3 Minimum breaking force of single wire and strand shall be as per IS: 2141 – 2000.
- 4.2.4 Type of coating of galvanise shall be heavy as per IS: 4826 – 1979 and the wires shall be galvanised after drawing, but without annealing.
- 4.2.5 Weight per metre length of the stranded stay wire shall be stated by the Tenderer.

5.0 TESTS AND TEST CERTIFICATES :

5.1 Test Report (To be submitted alongwith the Bid)

Copy of Test Report carried out within last five years on the items of material of the Tender in a NABL accredited Test House or Laboratory shall be submitted along with the tender as Pre-requisite. If there be any deficiency regarding non-submission of the Test Report, WBSEDCL reserves the right to cancel the tender unilaterally.

5.2 The following Acceptance Tests shall be carried out as per the relevant IS Codes.

5.2.1 Acceptance Tests for G.I. Wires according to IS: 280 –2006:

- a) Chemical Composition Test
- b) Verification of diameter
- c) Mechanical properties like Tensile Test, Wrapping Test, Bend Test etc.
- d) Coating Test for galvanise
- e) Test for conformity.

5.2.2 Acceptance Tests for G.I. Stay Wires according to IS: 2141–2000:

- a) Chemical Composition Test
- b) Ductility Test
- c) Verification of wire diameter
- d) Tensile and Elongation Test
- e) Galvanising Test
- f) Test for conformity.

6.0 MARKING:

Each coil shall be provided with a tag made of metal of suitable size securely attached on the inner part of the coil bearing the following information :

- A. Manufacturer's name or Trade Mark.
- B. Lot number and coil number.
- C. Quality and size of material.
- D. BIS Certification mark, if any.

7.0 TOLERANCE IN QUANTITY:

Tolerance in supply of total quantity will be allowed upto -2 % to +1% of the ordered quantity for each item.

8.0 PACKING:

Each coil of wire shall be suitably bound and fastened compactly.

9.0 WEIGHT OF COIL:

Weight of each coil shall be between 50 Kgs. & 100 Kgs.

10.0 PRICE :

The Price shall be FIRM.

11.0 Documents to be submitted at the time of physical delivery at consignee stores :

The following documents to be submitted by the venders to the Consignee Stores at the time of despatch to stores by the venders:-

- a) Copy of Purchase Order.
- b) Copy of Despatch Instruction
- c) Inspection Test Certificate
- d) Guarantee Certificate
- e) Proforma Invoice
- f) Calculation Sheet.
- g) Seal list and packing list
- h) Challan in triplicate
- i) Way bill, if applicable.